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Adsorption of hydrogen on nickel single crystal surfaces

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Adsorption of hydrogen on Ni(111), (100), and (110) surfaces was studied by means of LEED, energy loss spectroscopy, flash desorption, and work function measurements, as well as by the technique of laser-induced thermal desorption. Noticeable variations of the LEED intensities for Ni(111) and (100) indicate the formation of disordered adsorbed layers, whereas with Ni(110) the formation of "streaked" diffraction patterns and (at high coverages) of a 1×2 structure were observed. Hydrogen adsorption causes a strong damping of the intensity of the surface plasmon excitation and the appearance of an electron loss peak around 15 eV. Flash desorption experiments revealed for Ni(111) and Ni(100) the existence of β_1 and β_2 states, the former being only filled after completion of the β_2 state. Desorption from the β_2 state follows a second-order rate law for Ni(111) and Ni(100), but is of first order with Ni(110). Maximum increases of the work function $\Delta\varphi$ by 0.195, 0.170, and 0.530 eV were observed with Ni(111), (100), and (110), respectively. Adsorption isotherms $\Delta\varphi = f(p_{H_2})$ for Ni(111) and Ni(100) may be closely described up to medium coverages on the basis of a model that assumes second-order desorption and first-order adsorption kinetics. The validity of this model was further proved by measurements of the variation of coverage with time. The adsorption isotherms for Ni(110) have a unique shape indicating the occurrence of two-dimensional condensation due to attractive interactions in the adsorbed layer. Isosteric heats of adsorption at low coverages of 23 kcal/mole for Ni(111) and Ni(100) and of 21.5 kcal/mole for Ni(110) were evaluated. With Ni(110) the onset of condensation is characterized by an increase of E_{ad} by about 2 kcal/mole. A steplike decrease of E_{ad} at medium coverages accompanies the formation of the β_1 states.

I. INTRODUCTION

The interaction of hydrogen with nickel surfaces represents one of the most extensively investigated subjects among the studies on chemisorption. This is partly due to the catalytic properties of this metal in hydrogenation reactions, but the main reason lies probably in the fact that this system has been considered to be simple and therefore to be well suited for fundamental research on adsorption phenomena. Besides numerous papers on polycrystalline Ni surfaces,[1–15] a series of results with single crystalline samples using different techniques of investigation were also reported in the literature.[16–26] However, with the exception of a recent study by Taylor and Estrup[20] on Ni(110) all this work was performed without a control of the surface cleanness by means of Auger electron spectroscopy (AES). Nickel surfaces may readily interact with components of the residual gas atmosphere, in particular with carbon monoxide which may even decompose on Ni.[27,28] Spurious amounts of coadsorbed CO may markedly influence the adsorption properties for hydrogen[29] which might well explain some of the discrepancies existing in the literature.

The present paper describes the results of a series of experiments with the three most densely packed Ni planes (111), (100), and (110) obtained under stringent conditions of surface purity. The measurements were performed using Auger electron spectroscopy, low energy electron diffraction (LEED), electron energy loss spectroscopy (EELS), flash desorption, and laser induced thermal desorption techniques, as well as measurements of the variation of the electron work function.

II. EXPERIMENTAL

The experiments were performed using a stainless steel uhv system with a base pressure $<10^{-10}$ Torr. Details of the methods of investigation and of the sample preparation and cleaning procedures have been described elsewhere.[30,31] However, the following modifications should be mentioned:

(i) The technique of sample preparation by means of mechanical grinding and electropolishing was improved leading to LEED patterns from the clean surfaces of very high quality.

(ii) By using an additional titanium sublimation pump and prolonged exposure of the apparatus to hydrogen the partial pressure of CO was suppressed below $\sim 10^{-12}$ Torr and therefore this gas had no disturbing influence on the adsorption studies with hydrogen.

(iii) Oxidized tantalum was used as material for the reference electrode in the Kelvin contact potential measurements which could be performed with a noise level below 1 mV. The maximum work function change due to hydrogen adsorption was found to be highly sensitive to the state of cleanness of the surface.

(iv) Laser induced thermal desorption was achieved by focusing the light of a Nd pulse laser (pulse width 150 μsec, maximum energy 1.2 J) onto a small spot of about 1 mm^2 of the sample surface. The desorbing species was partly collected by a quadrupole mass spectrometer (Balzers QMG 111 A) mounted in line of sight to the surface. The partial pressure was recorded as a function of time using a storage oscilloscope or a digital averager. Details of this technique will be published elsewhere.[32] In particular it has to be noted that the magnitude of the pressure burst was found to be proportional to the adsorbed amount, provided that all the other parameters were kept constant.

III. RESULTS

A. LEED

1. Ni(111) and (100) surfaces

Germer *et al.*[16] reported in 1960 about LEED studies on hydrogen adsorption at Ni(111) and (100) surfaces where they did not detect any formation of additional diffraction spots.

The authors assumed the light hydrogen atoms to be only very weak scatterers and even if the periodicity of their arrangement would deviate from that of the substrate lattice this fact would prevent the appearance of "extra" LEED spots with considerable intensities. The present results agree with these earlier findings, as no variation of the geometry of the diffraction spots was observed even after H_2 exposures of more than 100 L (1 L $= 10^{-6}$ Torr · sec). However, the LEED intensities of the substrate spots and the background changed considerably under the influence of an adsorbed hydrogen layer.

These measurements were performed with a movable Faraday-cup collector. Figure 1 shows the intensity/voltage (I/V) profile for the (0, 0) beam of the clean Ni(111) surface (full line) and after hydrogen adsorption (dashed line). The angle of incidence of the primary electron beam deviated by 2° from the surface normal. The calculated positions of the primary Bragg peaks are indicated by arrows and the experimental peak positions are shifted towards lower energies due to the inner potential V_{in}. For the latter a value of $V_{in} = 11$ eV is derived. The I/V curve for the clean Ni(111) surface agrees with that measured by Ngoc *et al.*[33] and by Demuth and Rhodin.[34] The main features of this curve are contained in the theoretical results of Laramore.[35] From the positions of the primary Bragg peaks furthermore by means of a kinematical treatment the vertical distance between the topmost atomic layers may be evaluated. For the lattice constant normal to the surface a value of 3.52 ± 0.02 Å results which does not deviate within the limits of accuracy from the bulk lattice constant, which is again in agreement with the results of a dynamical calculation.[35]

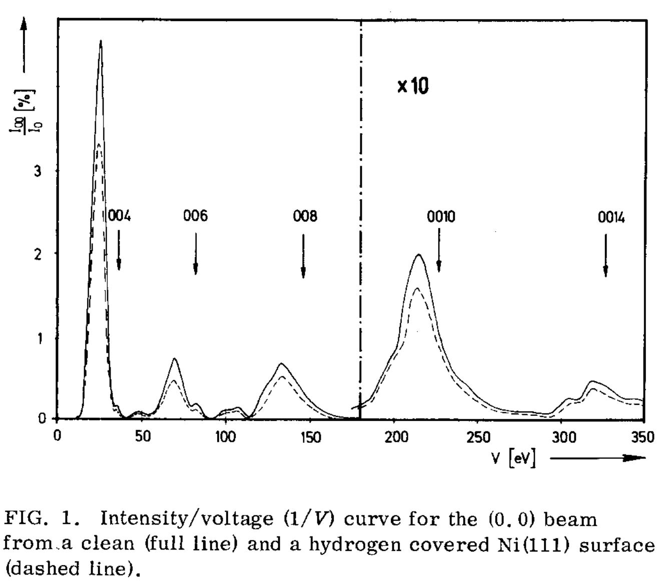

FIG. 1. Intensity/voltage (I/V) curve for the (0, 0) beam from a clean (full line) and a hydrogen covered Ni(111) surface (dashed line).

As can be seen from Fig. 1 adsorption of hydrogen causes a general decrease of the intensity of the specularly reflected beam without affecting the structure of the I/V profile. Simultaneously a small increase of the background intensity (i.e., outside the diffraction spots) was observed. It is concluded that at room temperature the adsorbed layer is disordered—in contrast to Pd(111) where characteristic variations of the I/V curves and a weak decrease of the background were observed which was attributed to the formation of a true 1×1 overlayer structure.[31] In any case an adsorbed hydrogen layer causes considerable variations of the LEED intensities which was predicted for example by Jennings and McRae[36] on the basis of a multiple-scattering model. The earlier concept[18,37] that noticeable variations of the LEED intensities after hydrogen adsorption should only occur on "open" planes in connection with a periodic reconstruction of the metallic surface atoms became soon the object of criticism[38] and is not supported by the present results.

Quite similar results were obtained with a Ni(100) surface which were already published elsewhere.[31] Again no formation of "extra" diffraction was observed after hydrogen adsorption at room temperature.

The I/V profile for the (0, 0) beam showed again the general decrease of the intensity upon H_2 adsorption. The inner potential was evaluated to $V_{in} = 12$ eV in agreement with previous measurements[39] and calculations.[35] For the lattice constant of the topmost layers normal to the surface a value of 3.52 Å was derived which again agrees with the bulk value within the limits of accuracy. The decrease of the intensities of the Bragg beams together with the observed increase of the background intensity is again ascribed to the formation of a (at room temperature) disordered adsorbed layer. [Probably ordered structures on Ni(100) and Ni(111) are formed at lower temperatures.] Furthermore there is no indication of a lattice expansion on both planes under the influence of hydrogen adsorption as was concluded for Pd(111).[31]

If the variation of the LEED intensities is observed as a function of hydrogen exposure one finds that the final stage is reached after about 5–10 L (1 L $= 10^{-6}$ Torr · sec) which indicates an initial sticking coefficient around 0.2.

2. The Ni(110) surface

In contrast to Ni(111) and Ni(100), on the (110) surface, hydrogen adsorption causes the appearance of new features in the LEED pattern, namely the formation of streaks between the substrate spots which finally coalesce into additional diffraction spots of a 1×2 structure.[17,18,20,23]

Figure 2 illustrates these variations of the LEED pattern. An H_2 exposure between 0.1 and 1 L causes the appearance of sharp streaks between the substrate spots in the [01] direction of the reciprocal lattice [Fig. 2(b)] which corresponds to the formation of a one-dimension-

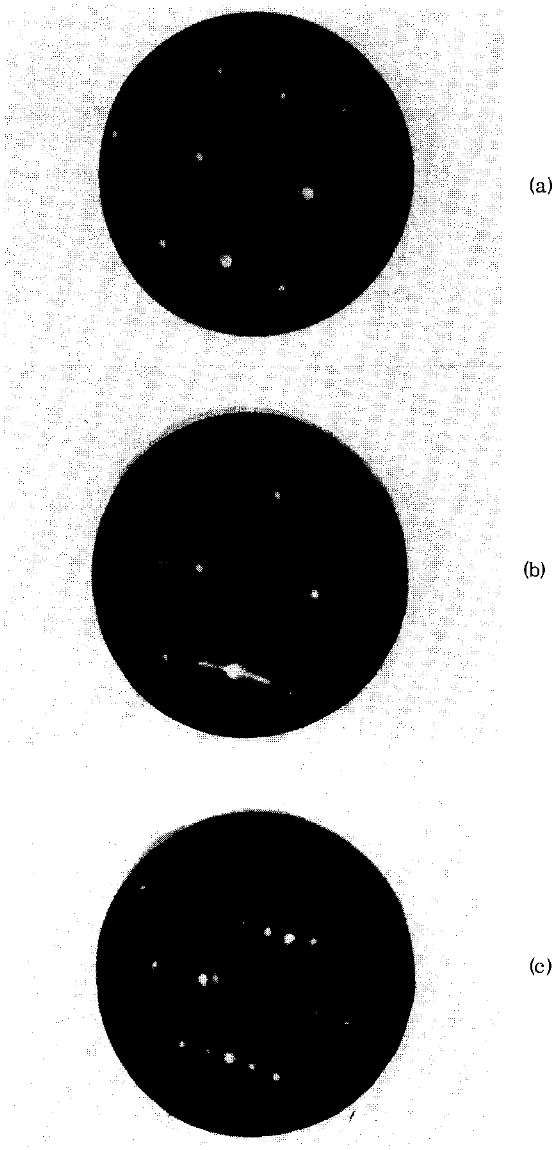

FIG. 2. LEED patterns from Ni(110). Primary energy E_p = 56 eV. (a) Clean surface. (b) Streak formation after H_2 exposure of about 0.5 L. (c) 1×2 structure at saturation of the adsorbate layer.

ally ordered arrangement of the scatterers in [10] direction of the real lattice, the periodicity in this direction being equal to that of the substrate lattice. Since the diffraction streaks are very sharp there must be a rather perfect long-range order in this direction over dimensions larger than the coherence width of the electron beam (~100 Å) whereas in the [01] direction there exists no order at this stage of adsorption. Since this type of LEED pattern is formed instantaneously at room temperature the adsorbed particles must have a rather high surface mobility. If the partial pressure of gaseous hydrogen is reduced the streaks in the diffraction pattern disappear within a short time from which it is concluded that hydrogen desorbs due to a moderate heat of desorption at this stage.

If the hydrogen exposure is increased beyond 1 L the streaks in the diffraction pattern contract continuously into a pattern of a 1×2 structure [Fig. 2(c)] connected with a steep increase of the coverage as becomes evident from the results of the contact potential measurements as discussed below. This 1×2 structure is stable after evacuation of the vacuum chamber even over hours which indicates that the desorption energy of this stage is higher than that for the "streaked" diffraction pattern. Similar observations were made in an earlier study[23] although the conclusions drawn then have to be revised to some extent.

In earlier experiments[17,18] the appearance of LEED "extra" spots of a 1×2 structure after hydrogen adsorption was considered as being caused by a reconstruction of the surface atoms. Although serious doubts about this interpretation arose in the meantime the present results are not able to rule out this possibility. Unfortunately no facilities to measure LEED intensities of a hydrogen covered Ni(110) surface were available at this stage of the experiments. May and Germer[17] reported on an I/V curve for the $(0, \frac{1}{2})$ "extra" spot which they considered as a support for their proposed model of surface reconstruction although of course the final decision of this question can only be expected on the basis of a dynamical analysis of the LEED intensities.

B. Electron energy loss spectroscopy (EELS)

Information on electronic interactions between the adsorbed particles and the substrate atoms may be obtained by measuring the energy distribution of backscattered electrons caused by bombarding the surface with a monoenergetic primary electron beam. Recently Küppers[40] demonstrated that chemisorption of CO, H_2, or O_2 on a Ni(110) surface may cause the appearance of new characteristic energy losses accompanied by variations of the "intrinsic" energy distribution in the range of about 30 eV below the elastic peak.

In the present experiments the LEED gun served as a source of primary electrons with an inherent energy spread of about 1%. The four-grids optics were used as a retarding field analyzer the amplitude of the modulation voltage being less than 0.5 V. Figure 3(a) shows the energy distribution for a clean Ni(100) surface taken at a primary energy $E_p = 74.5$ eV. The prominent features are three energy losses at 8, 18, and 26 eV below the primary peak which were already reported in the literature[41] and were interpreted as being caused by excitations of surface plasmons, bulk plasmons, and "combined" surface/bulk plasmons, respectively. Hydrogen adsorption [Fig. 3(b)] causes a drastic suppression of the intensity of the surface plasmon excitation whereas the two other peaks are nearly unaffected. However, there is some evidence for a new peak emerging around 15 eV which overlaps with the bulk plasmon excitation at 18 eV. The formation of a chemisorption induced energy loss at 15 eV after hydrogen adsorption was also discovered in the work of Küppers with the system H_2/Ni(110).[40] This author observed the slow formation of a second peak at 7.5 eV after prolonged interaction of hydrogen with the surface. However, there is some evidence that this feature was caused by coadsorption of spurious amounts of CO from the residual gas

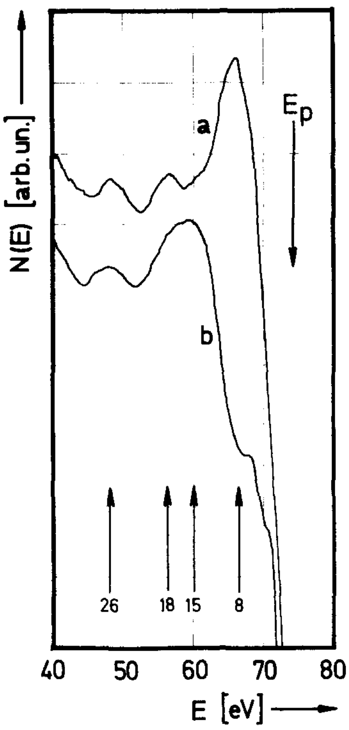

FIG. 3. Energy distribution of electrons backscattered from Ni(100). Primary energy $E_p = 74.5$ eV. Curve a: clean surface. Curve b: after hydrogen adsorption.

atmosphere. The energy loss at 15 eV is tentatively interpreted as being caused by the excitation of an electron from a filled chemisorption level into a higher (probably antibonding) unfilled state. This concept is strongly supported by measurements of the threshold energy for electron impact desorption. Lichtman *et al.*[18] found that electron bombardment of a hydrogen covered Ni(100) surface leads to the release of H^+, a process which was previously interpreted as being caused by the excitation of electrons from bonding into antibonding adsorption levels.[42] In the latter case a threshold energy of 15.0 eV was measured[19] which agrees with the energy of the characteristic loss.

With a clean and a hydrogen covered Ni(111) surface qualitatively the same features of the energy loss spectra were found as described above for Ni(100).

C. Flash desorption

If in a permanently pumped vacuum system the temperature of an adsorbate covered surface is continuously raised characteristic pressure maxima may be observed whose positions on the temperature scale depend, among other parameters, on the binding energy of that particular state of adsorption.[43,44] Extended measurements of this type were recently performed for example by Lapujoulade and Neil for H_2 on Ni(111),[24] Ni(100),[25] and Ni(110)[26] as well as by Wedler *et al.*[11] and by Völter and Procop[9] with polycrystalline Ni films, whose surfaces are mainly composed of (111) and (100) planes.[45] With heating rates < 1 K sec^{-1} Wedler *et al.*[11] reported for a completely covered surface the appearance of two desorption maxima around 310 (β_1) and 380 K (β_2 state). At small coverages at first only the β_2 state becomes filled. Similar observations were made by Völter and Procop.[9] Since the desorption temperature of the β_1 state is only slightly above room temperature hydrogen from this state is expected to desorb already during evacuation of the vacuum chamber if the sample is held at room temperature. Therefore some experiments with a Ni(111) surface were performed with an apparatus which allowed cooling of the sample down to $-15\,°C$.[46] Starting at this temperature with a heating rate of 3 K sec^{-1} desorption spectra of the type as shown in Fig. 4 were recorded. At small coverages only a single peak at about 370 K appears (β_2), and after complete filling of this state a second maximum around 330 K (β_1) is built up. These findings are in fairly good agreement with the results obtained with polycrystalline films[9,11] if the somewhat different heating speeds and the not very high accuracy of temperature determination with these experiments are taken into consideration.

The results described in the following were always obtained with starting temperatures at or above room temperature and therefore contain no information on the β_1 state on the above mentioned reasons.

A series of flash desorption spectra for Ni(100) taken at a constant heating rate of 15 K sec^{-1} and with different initial coverages is reproduced in Fig. 5. The peaks are shifting towards lower temperatures with increasing exposure which might be characteristic for a second order rate desorption process provided that the activation energy for desorption remains constant.

This latter assumption is supported by the equilibrium measurements described in the following sections. If desorption is a second-order rate process then the following relation between the activation energy for desorption E^* ($\approx$ adsorption energy E_{ad}) the frequency factor ν_2, the temperature T_m at which the pressure maximum occurs, the surface density of adsorbed particles n_s, and the heating rate b holds[44]:

$$E^*/RT_m^2 = n_s \nu_2 / b \cdot \exp(-E^*/RT_m)\,. \tag{1}$$

n_s is proportional to the area below the corresponding desorption peak. A plot of $\ln(n_s T_m^2)$ vs $1/T_m$ should yield a straight line which in the present case is rather well fulfilled.

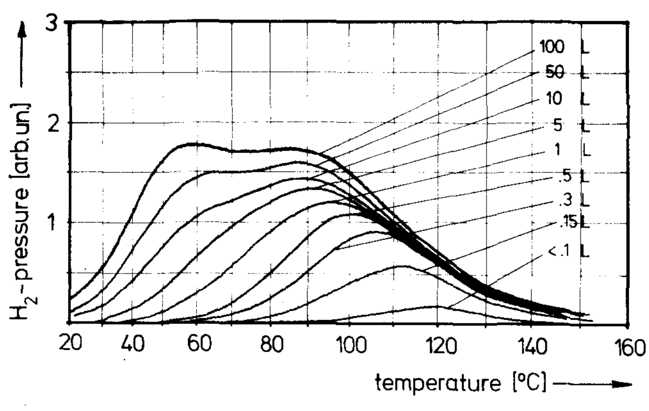

FIG. 4. Flash desorption curves for H_2/Ni(111). The exposure data are not very accurate since the pressure measurements were not performed in the vicinity of the sample with these experiments.

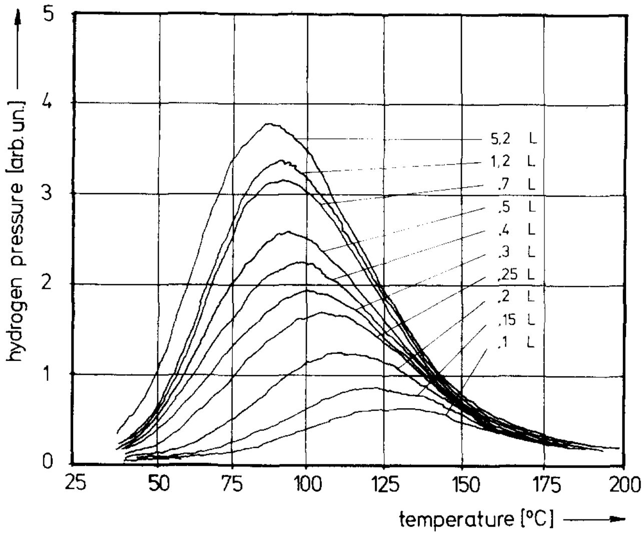

FIG. 5. Flash desorption spectra for the β_2 state of H_2/Ni(100).

In order to derive the activation energy for desorption E^* numerical values on the surface concentrations of adsorbed particles n_s and on the frequency factor ν_2 must be known. No attempt was made in the present study to evaluate n_s from a quantitative analysis of the desorbing amounts of H_2 due to the well-known difficulties in this respect with stainless steel uhv systems. For Ni(100) Lapujoulade and Neil[19] arrive at a maximum surface density of $n_{s,\max} = 3.3 \times 10^{14}$ cm^{-2} corresponding to $\theta_{\max} \approx 0.25$. This value appears to be rather small and will be discussed later in more detail. Nevertheless, it can easily be shown that an uncertainty for n_s within a factor of 10 causes a variation of E^* of less than 1 kcal/mole. The frequency factor ν_2 may be evaluated by recording flash desorption spectra with equal initial coverages but different heating rates b.[43] A value $\nu_2 = 8 \times 10^{-2}$ [cm^2 $particles^{-1} \cdot sec^{-1}$] was derived from such an analysis using heating rates between 4 and 40 K sec^{-1}. This value is in fairly good agreement with that of 2.5×10^{-1} obtained by Lapujoulade and Neil.[25] These authors arrived at an activation energy for desorption $E^* = 23.1 \pm 1$ kcal/mole which also agrees completely with the analysis of the present measurements yielding $E^* = 23 \pm 1$ kcal/mole.

The frequency factor ν_2 contains some information on the degree of mobility of the adsorbed hydrogen atoms. In the case of immobile adsorption $\nu_2 = 4 \times 10^{-1}$ [cm^2 $atoms^{-1} \cdot sec^{-1}$] is predicted whereas for a completely mobile two-dimensional gas a value of 4×10^{-3} would result.[25,47] The derived value of 8×10^{-2} indicates that the adsorbed hydrogen atoms are not immobile, but somewhat restricted in their surface mobility. This would agree with observations made in field emission experiments[21] whereas surface diffusion of hydrogen adsorbed on nickel starts at temperatures around 250 K.

Quite similar results, namely again second-order desorption kinetics with an activation energy of 23 kcal/mole for the β_2 state, were obtained from the spectra from Ni(111) which are therefore not outlined in detail here. Lapujoulade and Neil arrive at $E^* = 22.7$ kcal/mole and $\nu_2 = 0.2$ [cm^2 $atoms^{-1} \cdot sec^{-1}$].[24] The temperature for the β_1 state in Fig. 4 is about 40 °C below that of

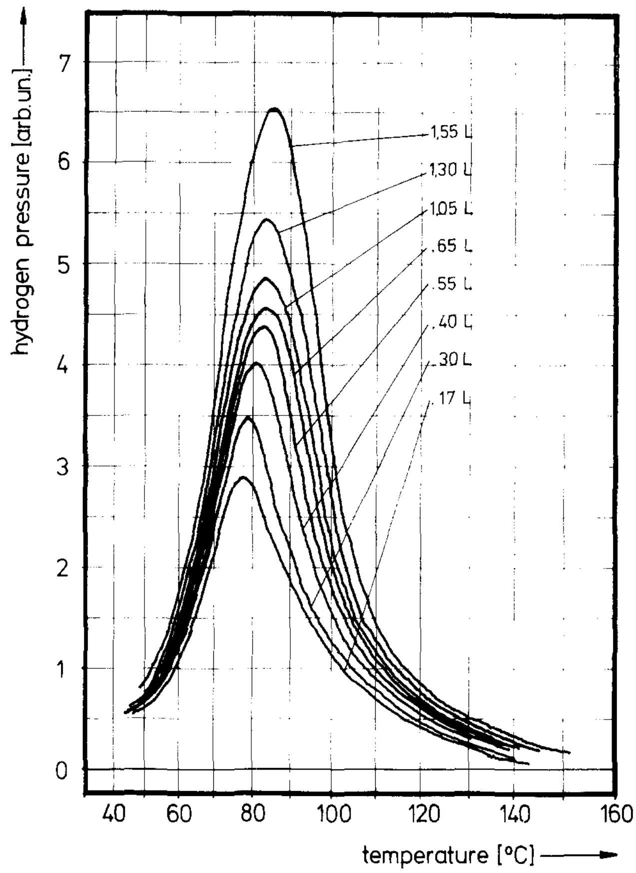

FIG. 6. Flash desorption spectra for the initial stages of H_2 adsorption on Ni(110).

the β_2 state from which an around 3 kcal/mole smaller value for the adsorption energy (i.e., ~20 kcal/mole) is estimated.

Striking differences as compared with the (111) and (100) planes were observed with Ni(110). A series of flash desorption spectra for different initial coverages is shown in Fig. 6. The temperature maxima are at first slightly shifted towards *higher* temperatures with

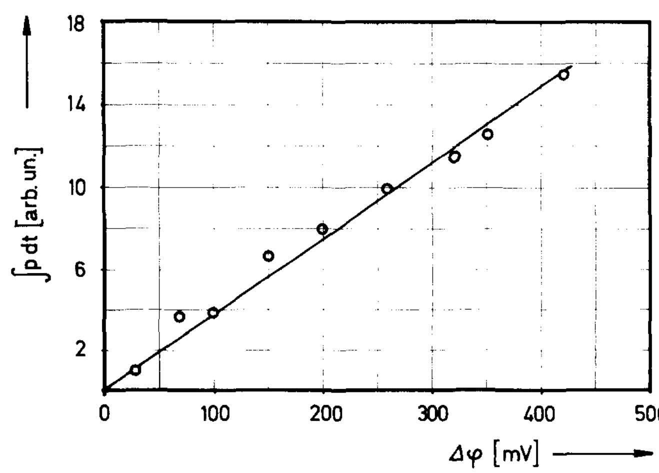

FIG. 7. Calibration of the work function change $\Delta\varphi$ for H_2/Ni(110) against the relative adsorbed amount as determined from the areas $\int p dt$ below the corresponding flash desorption curves.

increasing coverage and then remain practically constant over a wide range. The latter effect is characteristic for desorption processes obeying first order rate laws.[43] Assuming a reasonable value of $\nu_1 = 10^{13}$ sec^{-1} for the first order frequency factor again a value of 23 kcal/mole for the activation energy of desorption is obtained. The observed initial increase of T_m with increasing coverage might be caused either by a zero order desorption process (which seems physically unreasonable) or by the fact that in this range of coverage the adsorption energy increases with increasing θ. Lapujoulade and Neil[26] arrived at similar conclusions from their flash desorption measurements, namely a value of 20.3 kcal/mole at very low surface concentrations ($< 10^{13}$ cm^{-2}) which increases to about 23 kcal/mole. Further elucidation of this problem will be obtained from the adsorption isotherms.

D. Contact potential variation and coverage

Chemisorption of hydrogen on nickel causes an increase of the work function, that means the adsorbed particles become negatively polarized. The maximum change found with polycrystalline Ni surfaces lies in the range of about 0.35 eV.[48,49]

In a work with Ni(110) (which we now believe was not performed under perfectly clean surface conditions)[23] a value of $\Delta\varphi_{max} = 0.42$ eV was observed whereas recently Taylor and Estrup[20] with the same plane found a somewhat higher value of 0.55 eV. Earlier field emission work[21] indicates that hydrogen adsorption on the Ni(110) plane causes a higher increase of the work function than at the surfaces with other orientations.

Most of the present measurements were performed with a self-compensating Kelvin method[23] using oxidized tantalum as material for the reference electrode. A series of control measurements were made by the diode method using the LEED gun as cathode. The results of both techniques agreed always to within a few meV which demonstrates the inertness of the reference electrode under the chosen experimental conditions. The measured maximum increases, $\Delta\varphi_{max}$ at room temperature and pressures up to 10^{-4} Torr for the three single crystal surfaces were as follows:

Plane	$\Delta\varphi_{max}$(eV)
Ni(111)	0.195
Ni(100)	0.170
Ni(110)	0.530.

Attempts were made to obtain a calibration of $\Delta\varphi$ against the coverage by measuring the areas below the corresponding flash desorption curves $\int p dt$. The latter quantity is directly proportional to the adsorbed amount. Figure 7 shows the results for the Ni(110) surface which demonstrate that at least up to $\Delta\varphi = 0.42$ eV, i.e., about 80% of $\Delta\varphi_{max}$, the variation of the work function is proportional to the coverage. The same conclusions were drawn in an earlier paper.[23] No further data are available for even higher coverages since the adsorbed gas desorbs at room temperature upon evacuation of the vacuum system before the flash desorption experiments can be started (cf. Sec. III C). This difficulty was also present for the Ni(100) and Ni(111) surfaces. In both cases again $\Delta\varphi$ was shown to be strictly proportional to the coverage up to about 50% of $\Delta\varphi_{max}$. This corre-

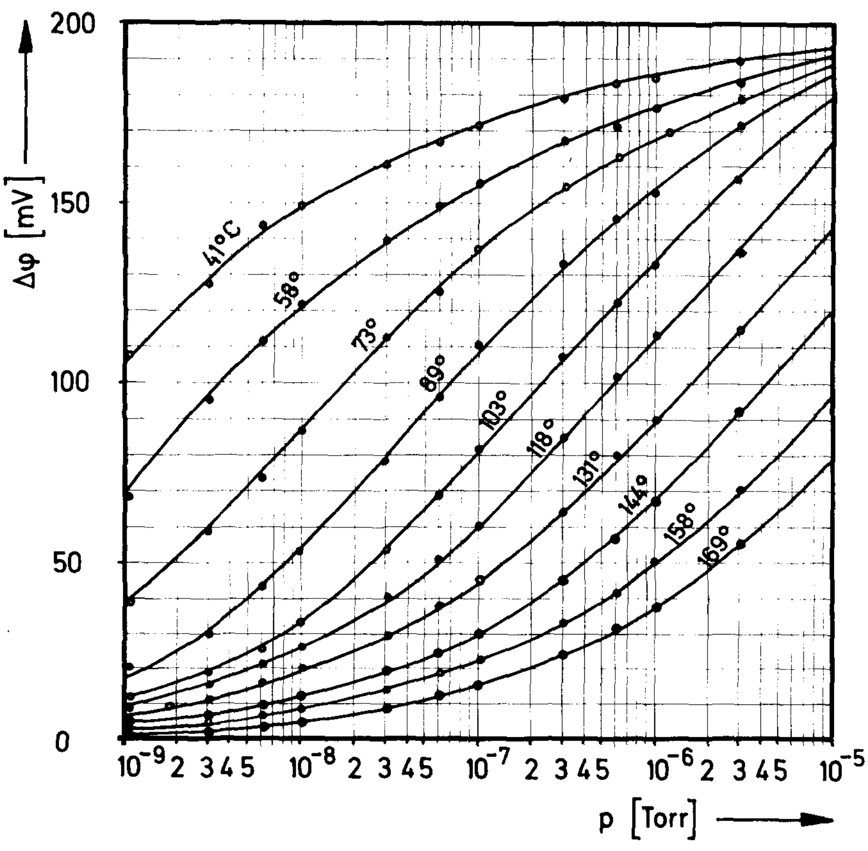

FIG. 8. Adsorption isotherms $\Delta\varphi$ vs pressure for H_2/Ni(111).

sponds to the range of coverages where the β_2 state becomes completely filled. Measurements with polycrystalline Ni films[12,49] indicate that in the range of the β_1 state (i.e., above half the saturation coverage) the slope of the $\Delta\varphi$ vs θ curve becomes smaller which very probably is also the case with the (111) and (100) single crystal surfaces.

Exact data on the absolute quantities of chemisorbed hydrogen atoms at saturation coverage are unfortunately not yet available. Lapujoulade and Neil arrive at values of 3×10^{14} atoms/cm^2 ($\theta_{max}\approx0.15$), 3.3×10^{14} atoms/cm^2 ($\theta_{max}\approx0.25$), and 3.6×10^{14} atoms/cm^2 ($\theta_{max}\approx0.4$) for Ni(111), Ni(100), and Ni(110), respectively.[24-26] These numbers appear to be rather small and are in disagreement with other results. Volumetric measurements with Ni(111) in an uhv glass system yielded recently $n_{s,max}\approx1.3\cdot10^{15}$ cm^{-2} or $\theta_{max}\approx0.7$ at room temperature.[50] This agrees with some preliminary (and not very accurate) measurements at the beginning of the present studies.[46] With polycrystalline Ni surfaces several groups[6,15] arrived at values for $n_{s,max}$ around 1.5×10^{15} cm^{-2} which also fits into this picture.

A high degree of uncertainty is present with regard to the maximum hydrogen coverage on Ni(110). May and Germer[17] estimated from flash desorption measurements from a Ni(110) surface exhibiting the 1×2 LEED pattern θ_{max} to be in the range between 1.6 and 2.2. Recent coadsorption studies of H_2 and CO on Pd(110) lead to the conclusion that the H − 1×2 structure corresponds either to $\theta=1$ or to $\theta=2$.[51] Since the same LEED pattern is formed with Ni(110) this result can be assumed to be also valid for this plane, that means $n_{s,max}=1.16\times10^{15}$ or 2.23×10^{15} cm^{-2}. Accurate measurements on absolute adsorbed quantities are badly needed.

E. Adsorption isotherms and isosteric heats of adsorption

The adsorption of hydrogen on nickel is completely reversible above room temperature.

As a consequence at a given temperature and pressure the adsorption/desorption equilibrium between the gas phase and the adsorbed particles leads to a certain coverage θ for which in the present work the contact potential change $\Delta\varphi$ was taken as a measure. From a series of such adsorption isotherms the isosteric heat of adsorption may be derived by using the Clausius–Clapeyron equation:

$$\left(\frac{d\ln p}{d(1/T)}\right)_{\theta=\mathrm{const}} = -\frac{E_{ad}}{R}\,. \qquad (2)$$

Figure 8 shows a series of adsorption isotherms $\Delta\varphi = f(p_{H_2})$ at different temperatures for Ni(111). The corresponding results for Ni(100) are reproduced in Fig. 9. In order to derive E_{ad} by means of Eq. (2), we must take points at constant coverage from different isotherms. If $\Delta\varphi$ is used as a measure for θ it is a necessary prerequisite that at constant coverage $\Delta\varphi$ is independent of temperature. Although there is no physical reason why the dipole moment of an adsorbate complex should vary considerably with temperature care has to be taken on the change of the work function of the metal itself. This effect was found to exist as a decrease of the work function of clean nickel surfaces by about 1.7×10^{-4} [eV/K]. Similar observations with nickel[52,53,27] as well as with

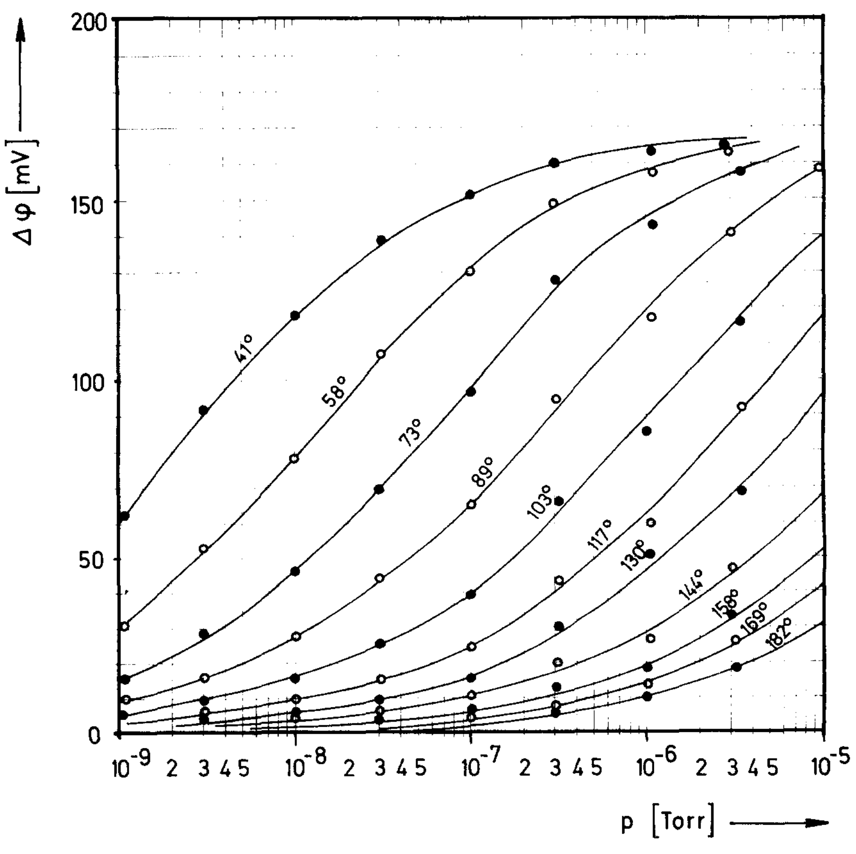

FIG. 9. Adsorption isotherms $\Delta\varphi$ vs pressure for H_2/Ni(100).

other metals[54,55] are reported in the literature. Although this effect was included in the derivation of E_{ad} it causes a considerable degree of uncertainty at very low coverages corresponding to $\Delta\varphi \leq 40$ meV, since $\Delta\varphi(T)$ is unknown for the adsorbate covered surfaces.

Figure 10 shows the so-derived isosteric heat of adsorption E_{ad} as a function of $\Delta\varphi$ for Ni(111). The results for Ni(100) are drawn in Fig. 11. In both cases up to about $\Delta\varphi \approx 100$ meV E_{ad} is constant (22.6 and 22.7 ± 0.5 kcal/mole, respectively). In this range $\Delta\varphi$ was shown to be directly proportional to the coverage as discussed explicitely in the foregoing section. This range obviously corresponds to the β_2 state identified in the flash desorption experiments, and also the binding energies are in complete agreement with the results of this type of experiments. In both figures the values for E_{ad} derived for $\Delta\varphi < 40$ meV are omitted. The evaluation of the adsorption isotherms at even smaller coverages indicates in both cases an increase of E_{ad} with decreasing coverage up to ~27 kcal/mole. Although in a series of papers with polycrystalline Ni surfaces[1,8,50] such an increase is reported, with the present single crystal results we have no further evidence (e.g., from the flash desorption spectra or from the shape of the adsorption isotherms which will be discussed later) for such an effect and therefore consider this context as an artifact caused by the uncertainty of the variation of the work function with temperature of the hydrogen covered surfaces. Recent volumetric measurements indicate that for Ni(111)/H_2 E_{ad} is constant down to a few percent of saturation coverage.[50]

Around $\Delta\varphi = 100$ meV E_{ad} drops steplike by about 1.5 kcal/mole to a new constant value and then finally decreases continuously when the coverage approaches saturation.

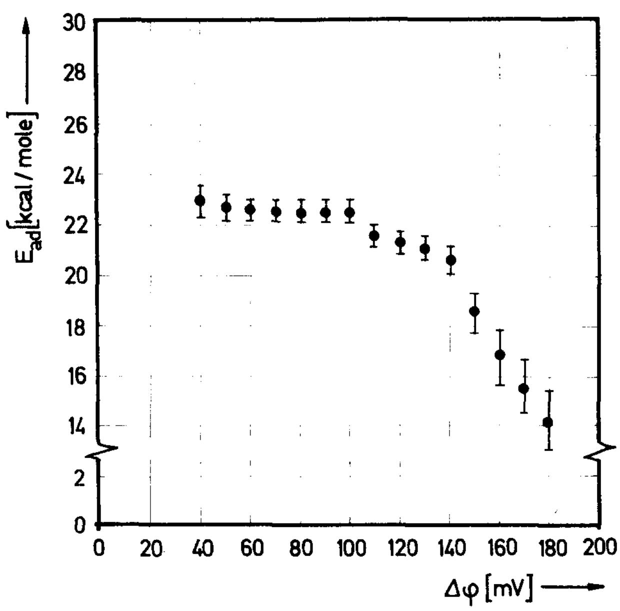

FIG. 10. Isosteric heats of hydrogen adsorption on Ni(111) as a function of the work function increase $\Delta\varphi$.

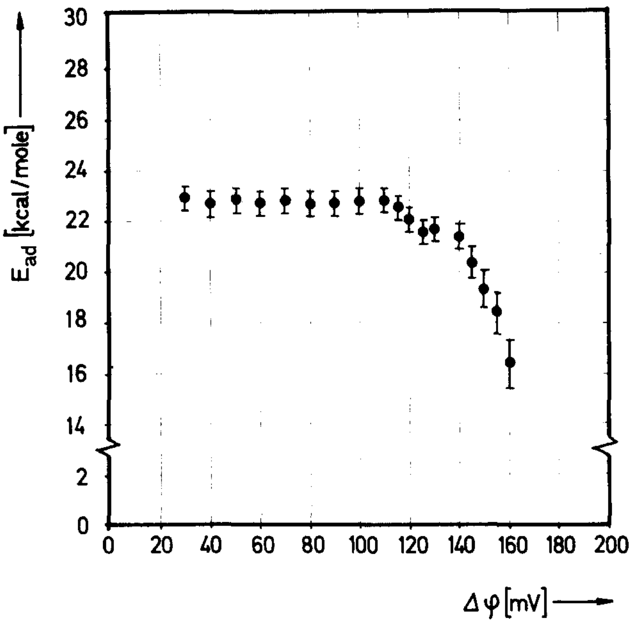

FIG. 11. Isosteric heats of H_2 adsorption on Ni(100) as a function of $\Delta\varphi$.

This second range has to be identified with the β_1 state found in the flash desorption experiments. Probably the proportionality between $\Delta\varphi$ and θ no longer holds in this range but the E_{ad} vs $\Delta\varphi$ curve is to be spread somewhat in order to obtain the E_{ad} vs θ correlation. Whereas Ni(111) and (100) are very similar in this context the (110) surface exhibits again a different behavior. The adsorption isotherms measured with this plane are reproduced in Fig. 12. At a given temperature the coverage rises very steeply as soon as a certain pressure is reached. Isotherms of this type are typical for two-dimensional condensation phenomena and were frequently discussed in the literature in connection with physical adsorption where the van der Waals attraction forces may cause such phase transitions.[56,57] In the case of chemisorption on metals to our knowledge such an effect has never been observed before. It was found that the shape of these isotherms is very sensitively influenced by trace amounts of surface impurities, in particular carbon monoxide.[29] This is probably also the reason why we failed to observe such an effect in an earlier study of this system.[23]

The isosteric heat of adsorption for Ni(110) as a function of $\Delta\varphi$, as derived from the adsorption isotherms, is plotted in Fig. 13. At very low coverages E_{ad} seems to have a constant value of ~21.5 kcal/mole and then increases at about $\Delta\varphi = 100$ meV to a new constant value of 23.4 kcal/mole. This increase of E_{ad} coincides with the onset of condensation as becomes evident from the adsorption isotherms. Also the shift of the flash desorption maxima towards somewhat higher temperatures observed in this range of coverage is in agreement with these findings and again reflects an increasing binding energy due to the operation of attractive interactions between the adsorbed particles.

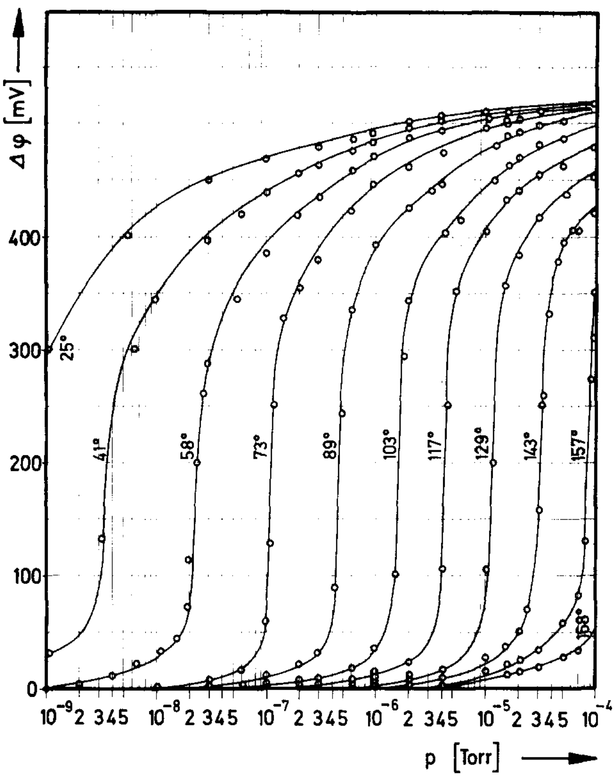

FIG. 12. Adsorption isotherms $\Delta\varphi$ vs pressure of H_2/Ni(110).

At $\Delta\varphi = 300$ meV E_{ad} again decreases stepwise by ~3 kcal/mole as observed with the other two planes and therefore again the formation of a β_1 state is concluded, although no flash desorption data for this state on Ni(110) are available.

F. Adsorption kinetics

The rate of adsorption is determined by the rate of impingement of H_2 molecules from the gas phase $\dot{N}$ and by the sticking coefficient s. $\dot{N}$ follows from the kinetic theory of gases

$$\dot{N} = \frac{1}{\sqrt{2\pi MRT}} \cdot p_{H_2} = \gamma \cdot p_{H_2} \left[\frac{\text{molecules}}{\text{cm}^2 \text{ sec}}\right] \tag{3}$$

(M is the molecular weight; p_{H_2} is the hydrogen pressure in torr). s may be written in terms of the initial sticking coefficient at zero coverage s_0:

$$s = s_0 \cdot f(\theta) . \tag{4}$$

In the case of reversible adsorption the variation of the density of adsorbed particles n_s with time may then be written as

$$\begin{aligned} dn_s/dt &= s\dot{N} - \delta g(\theta) \\ &= \gamma p_{H_2} s_0 f(\theta) - \delta g(\theta) . \end{aligned} \tag{5}$$

The first term describes the rate of adsorption and the second the rate of desorption, δ being the rate constant for desorption and the function $g(\theta)$ containing the reaction order of this process. If desorption can be neglected $s(\theta)$ therefore may be evaluated from the slope $dn_s/dt \propto d\theta/dt$ of the variation of coverage with time, as is usually done. In the case of reversible adsorption sometimes again this slope is used for a determination of a formal sticking coefficient, which however is felt to be without great physical meaning since it includes also the effect of desorption.

Of primary interest in understanding the mechanism of adsorption kinetics is the function $f(\theta)$. Since hydrogen adsorption on nickel is reversible above room temperature Eq. (5) will be used in the following analysis of experimental results in order to obtain informations about $f(\theta)$ by means of a trial-and-error procedure. Experiments to obtain the variation of the relative coverage with time were performed in detail for H_2/Ni(100) by using the laser-induced thermal desorption technique as a monitor for the surface concentration of adsorbed particles.

Figure 14 shows a series of such "covering" curves $\theta_{rel}(t)$ for H_2/Ni(100) at room temperature and at different H_2 pressures, starting with clean surfaces at $t=0$. In order to correlate Eq. (4) with the experimental results the functions $f(\theta)$ and $g(\theta)$ have to be inserted. From the flash desorption experiments we know that desorption follows a second order rate law, i.e., $g(\theta) = \theta^2$. The rate of desorption is then given by

$$v_{des} = \delta\ \theta^2 = \nu_2\ n^2_{s,\max} e^{-E^*/RT}\ \theta^2 , \tag{6}$$

where ν_2 is the frequency factor for second-order desorption, n_{max} the maximum density of adsorbed particle per cm^2, and E^* the activation energy for desorption. The simplest form for $f(\theta)$ is to assume that adsorption is a first order rate process with respect to the uncovered portion of the surface, i.e., $f(\theta) = 1 - \theta$. This implies a linear decrease of the sticking coefficient with increasing coverage, as it was indeed observed with polycrystalline Ni films.[6,7] The integrated form of Eq. (5) is then given by

$$\theta(t) = K(p,\ T)\frac{1 - \exp(-rt)}{1 - D\exp(-rt)}, \tag{7}$$

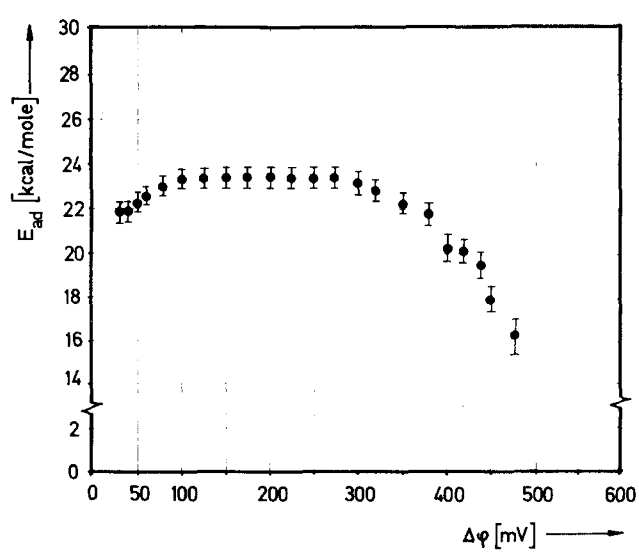

FIG. 13. Isosteric heats of H_2 adsorption on Ni(110) as a function of $\Delta\varphi$.

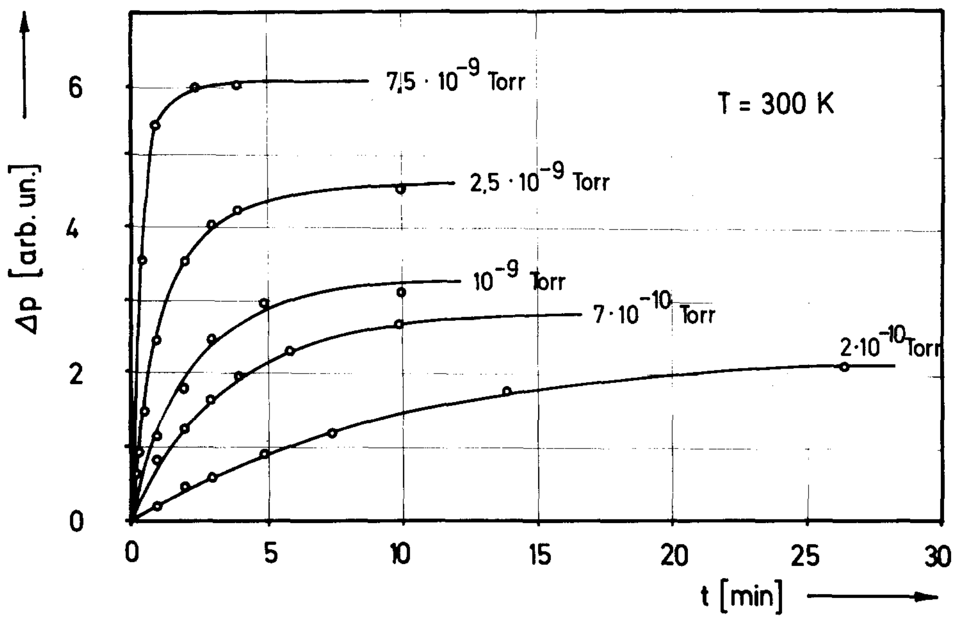

FIG. 14. Variation of the relative hydrogen coverage on Ni(100) with time at 300 K and at different H_2 pressures, starting at $t = 0$ with clean surfaces. The adsorbed amounts were determined by the pressure increases Δp due to laser induced thermal desorption.

where

$$r = \sqrt{\gamma^2 s_0^2 p_{H_2}^2 + 4\gamma s_0 \delta p_{H_2}},$$

$$K(p, T) = (r - \gamma s_0 p_{H_2})/2\delta,$$

$$D = (\gamma s_0 p_{H_2} - r)/(\gamma s_0 p_{H_2} + r).$$

Rearranging Eq. (7) yields

$$r = \ln \frac{1 - (\theta D/K)}{1 - (\theta/K)}. \tag{8}$$

If this equation is valid, a plot of $\ln\{[1 - (\theta/K)D] \times (1 - \theta/K)^{-1}\}$ vs the time t should give a straight line with slope r. Figure 15 shows that this is fulfilled for a series of hydrogen pressures and thereby it is indeed proven that the sticking coefficient for H_2 on Ni(100) obeys the simple law

$$s = s_0(1 - \theta). \tag{9}$$

Only rough estimates for the numerical value of the initial sticking coefficient s_0 to be about 0.25 can be made on the basis of the present results which is again partly related to the uncertainty connected with the absolute coverage. With polycrystalline Ni surfaces, values between 0.2 and 0.4[4,6,7,12] are reported whereas Lapujoulade and Neil found with Ni(111) and Ni(100) a much smaller value of about 0.1.[24,25] Detailed measurements on the adsorption kinetics at Ni(110) are not yet available, but there is strong evidence that s_0 is in the same range of magnitude as for Ni(111) and Ni(100). The previously observed slow transformation of the adsorbed state into a more tightly bond one[23] could not be reproduced under the present stringent conditions of surface cleanness. It is therefore concluded that this effect was caused by spurious amounts of impurities, probably originating from the residual gas atmosphere.

IV. DISCUSSION

A. Ni(111) and Ni(100)

These two planes showed nearly identical behavior and thus will be discussed together. The value of 23 kcal/mole found for the adsorption energy at low and medium coverages agrees well with a series of recent studies with single crystals[24,25] and with polycrystalline nickel, although in the latter cases there exists some disagreement with the results of Wedler's group who determined calorimetrically a somewhat lower value of ~18 kcal/mole. The independence of E_{ad} on coverage up to complete filling of the β_2 state indicates that there exist no (or only negligible) long-range interactions between the adsorbed H atoms. From the derived simple rate laws for adsorption and desorption kinetics the theoretical adsorption isotherms may be derived, namely simply

$$p = C^{-1}[\theta^2/(1 - \theta)], \tag{10}$$

where

$$C = \frac{s_0}{\nu n_{s,\max}^2 \sqrt{2\pi M R T}} e^{E^*/RT}.$$

Figure 16 shows some of the experimental data for Ni(111) together with the theoretically derived isotherms which were adjusted to one experimental point.

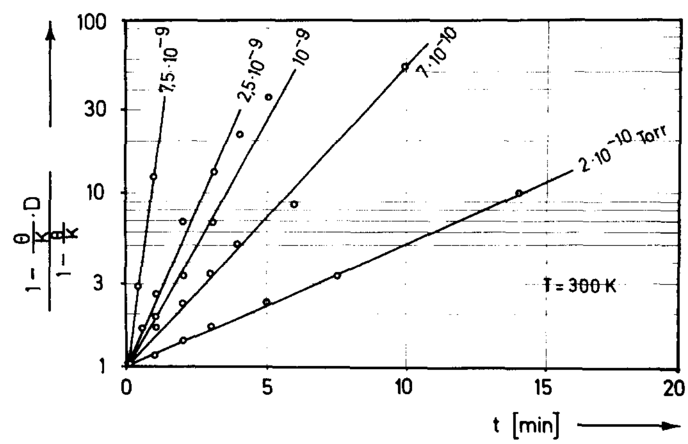

FIG. 15. Evaluation of the experimental data of Fig. 14 with Eq. (8).

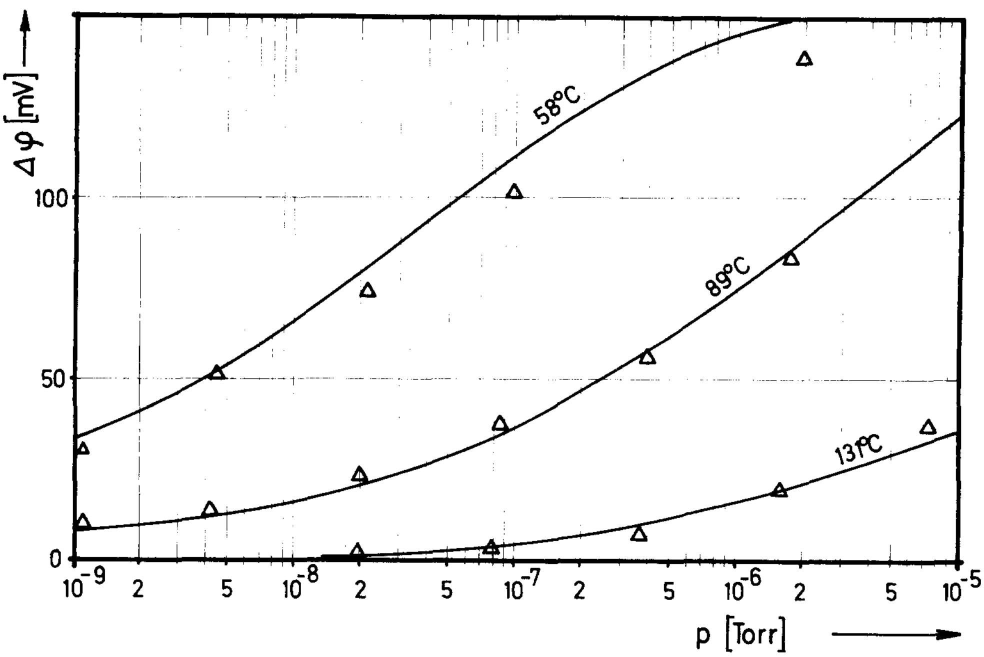

FIG. 16. Comparison of the theoretical adsorption isotherms of Eq. (10) with some of the experimental data for Ni(111).

In order to compare isotherms for different temperatures $E^* \approx E_{ad} = 23$ kcal/mole was taken, assuming that the rate of adsorption is a nonactivated process. The agreement between theoretical and experimental data is rather satisfying up to about half the maximum value of $\Delta\varphi$. The assumption underlying Eq. (10) that desorption is a second-order rate process leads to the prediction that at low coverages $\theta \propto \sqrt{p_{H_2}}$, which is also in very good agreement with the experimental results and was previously already found for hydrogen adsorption on palladium.[58] The general picture is therefore the following: H_2 adsorbs dissociatively, the recombination of two adsorbed H atoms being the rate-determining step for desorption. The adsorbed layer forms no ordered structure (as indicated from the LEED data) and a high surface mobility of the adsorbed H atoms may be assumed.

The β_1 state forms after completion of the β_2 state and is characterized by a decrease of the adsorption energy by about 2 kcal/mole. Similar observations have been made for example with hydrogen adsorption on tungsten,[59,60] where this transformation was found to be associated with significant variations of the electronic properties of the adsorbate complex.[60,61] Since the β_1 state on nickel desorbs already at room temperature no detailed results are available from the present studies.

However, for example, measurements of the variation of the electrical resistance of thin Ni films as a function of hydrogen coverage[15] indicate that also with this system the transformation from the β_2 into the β_1 state is associated with considerable changes of the electronic properties. A possible explanation for this effect would be the existence of two different types of adsorption sites. However, a statistical treatment by Toya[62] revealed that the observed shape of the flash desorption spectra as well as the stepwise decrease in the $E_{ad}(\theta)$ curve at medium coverages might also be the result of repulsive interactions between neighboring adsorbed particles. An interesting type of interaction between adsorbed particles takes place indirectly via the metallic electrons and may have even oscillatory character.[63] Recently this "through-bond" interaction between two H atoms on W(100) has been calculated by Grimley and Torrini[64] who found a short range repulsion with an energy of 3–6 kcal/mole, which is at least in the right order of magnitude. It is therefore concluded that the differences between β_1 and β_2 states on Ni surfaces are caused by such a short-range interactional effect.

B. Ni(100)

Although the average binding energy of hydrogen on Ni(110) appears to be quite similar to those determined for the other planes there exist obviously completely different interactions between the adsorbed particles leading to the unusual behavior of this surface.

Isotherms of the type as reproduced in Fig. 12 were discussed frequently in the literature on physical adsorption[56,57] and are the analogs to the van der Waals equation of state for real gases. The reason for the observed condensation phenomenon, i.e., the sudden increase of the coverage above a critical gas pressure, must be sought in the existence of attractive forces between the adsorbed particles. These attractive interactions must be operating in both crystallographic orientations on the surface, since only one-dimensional attractions are not able to cause such a phase transition.[65] The following picture arises for hydrogen adsorption at Ni(110): At very low coverages the adsorbed hydrogen atoms are randomly distributed over the surface with a large mean distance between neighboring particles. The adsorption energy (cf. Fig. 13) is about 21 kcal/mole in this range and therefore somewhat smaller than on Ni(111) and Ni(100). At a coverage corresponding to $\Delta\varphi = 60$ meV the attractive forces in [10] direction cause the formation of one-dimensional chain-like associates with a periodicity $a_0 = 2.49$ Å leading to the formation of the observed streaks in the LEED pattern. This onset of condensation is accompanied by an increase of the adsorption energy as becomes evident from the flash desorption data and from the isosteric heat of adsorption measurements. The interaction in [01] direction is much weaker since at first no periodicity in this direction is observed. Only a further increase of the coverage leads to continuous formation of long-range order in this direction whereas the adsorption energy remains practically constant in this range of coverages. For the 1×2 structure the length of periodicity in [01] direction is ~7 Å indicating the long-range character of this type of interaction.

No structure model is proposed for the 1×2 structure for the following reasons: Unequivocal information on the surface concentration of hydrogen adsorbed on Ni(110) is not yet available, but there is strong evidence that the unit cell of the 1×2 structure contains more than a single adsorbed particle. The formation of the 1×2 LEED pattern was formerly considered as being caused by a rearrangement of the surface atoms.[17,18] A possible explanation for the observed attractive interactions between adsorbed H atoms might then be found in a similar manner as for hydrogen dissolved in Pd, where the attraction is ascribed in terms of a compensation of the lattice strains.[66] On the other hand no indication for a condensation effect as observed with H_2/Ni(110) was found in the case of hydrogen adsorption on Pd(110) although the same 1×2 LEED pattern was formed.[58] The high $\Delta\varphi$ value for H_2/Ni(110) is a further argument against the picture that the hydrogen atoms are somewhat imbedded into the Ni surface atoms. Considerable variations of the LEED intensities after hydrogen adsorption were not only observed with Ni(110) but also with the other nickel planes and may be explained by multiple scattering effects at the surface as treated theoretically by McRae and Jennings.[36] Therefore the present results yield no support for the "reconstruction" model although its validity cannot be completely ruled out.

If the binding energies of hydrogen on the three different Ni single crystals planes are compared with each other only very small differences are found just as predicted by a theoretical treatment based on the extended Hückel method.[67] This is obviously also the reason why results obtained with clean polycrystalline Ni surfaces have much in common with those found with single crystals. The H_2/Ni system is again an example for which the "structural" factor is only of minor relevance similar as in the case of CO/Ni,[27,30] H_2/Pd,[57] CO/Pd.[68]

V. SUMMARY

Hydrogen adsorption at room temperature on Ni(111) and Ni(100) causes no variation of the LEED spot patterns but does induce noticeable changes of the LEED intensities, from which the formation of disordered adsorbed layers is concluded. In the case of Ni(110) with increasing coverage the well-known formation of streaks finally coalescing into "extra" spots of a 1×2 structure was observed.

Measurements of the energy distributions of backscattered electrons revealed a strong damping of the surface plasmon excitation and the occurrence of a loss peak at ~15 eV which is tentatively ascribed to an excitation within the system of chemisorption-induced electronic states.

Flash desorption experiments showed for Ni(111) and Ni(100) the existence of two states (β_1 and β_2) with desorption temperatures around 330 and 370 K, the β_1 state only being formed after complete filling of the β_2 state. Desorption from the β_2 state follows a second-order rate law, and the frequency factor was determined as $\nu = 8 \times 10^{-2}$ [$\mathrm{cm}^2\ \mathrm{atoms}^{-1}\cdot\mathrm{sec}^{-1}$]. With Ni(110) desorption from the β_2 state is a first-order rate process. At low coverages the temperature maxima of the flash desorption peaks are shifted to somewhat higher values with increasing coverage, indicating a slight increase of the adsorption energy.

Maximum increases of the work function by 0.195, 0.170, and 0.530 eV were observed with Ni(111), (100), and (110), respectively. $\Delta\varphi$ is proportional to the coverage θ at least up to $\frac{1}{2}\Delta\varphi_{max}$. Measuring $\Delta\varphi$ at different temperatures and hydrogen pressures lead to plots of adsorption isotherms. For Ni(111) and Ni(100) these may up to medium coverages very closely be described by theoretical isotherms derived from a model assuming second-order desorption and first-order adsorption, i.e., the sticking coefficient being proportional to $(1-\theta)$. The validity of the latter assumption was checked by an analysis of $\theta(t)$ curves obtained by the laser-induced thermal desorption technique.

The adsorption isotherms for Ni(110) have a unique shape indicating the occurrence of two-dimensional condensation due to the operation of attractive interactions between the adsorbed particles.

Isosteric heats of adsorption were derived from the adsorption isotherms. Values of ~23 kcal/mole for Ni(111) and Ni(100) and of ~21.5 kcal/mole for Ni(110) were evaluated at small coverages. In the latter case the onset of condensation is coupled with an increase of E_{ad} by ~2 kcal/mole. With all three planes the adsorption energy then remains constant up to medium coverages and then decreases steplike by about 2 kcal/mole, characterizing the formation of the β_1 state.

ACKNOWLEDGMENTS

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